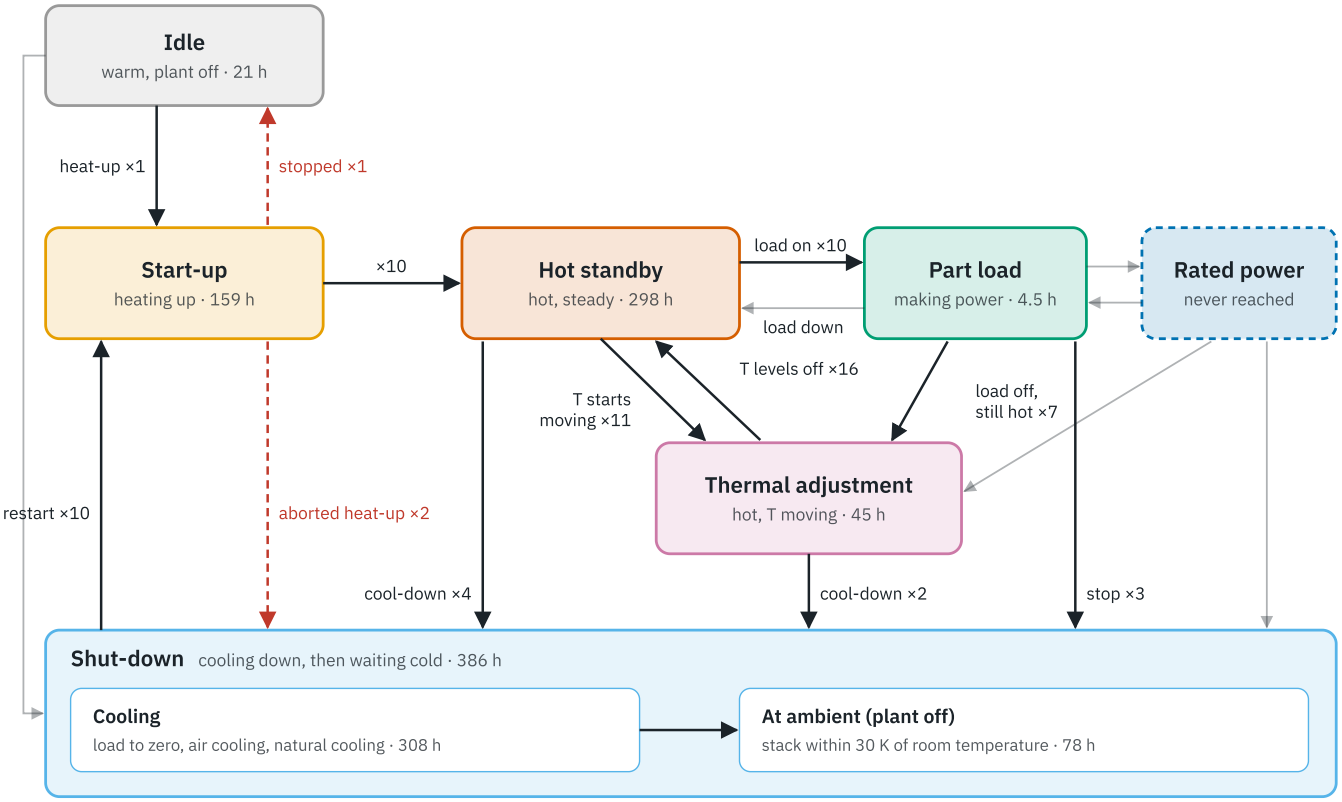
SOFC POC · 39 DAYS OF PLANT DATA · 2025-10-30 TO 2026-03-30

SOFC Stage Map

The plant is always in one of seven stages. The arrows below are the only moves the stage labels allow. Each count is how often that move happened in the 39 days of data.

Play Step Reset

Press Play to follow 17–20 March 2026 through the map, or Step to go one move at a time.



allowed, seen in the data (×count) allowed, not seen yet seen, outside the design

Hours are time spent in each stage, out of 914 h of labelled data (90 episodes). Rated power is drawn dashed because the plant never reached it. Source: Synthetic data generation/library (transitions.csv, episodes.csv), built from data_analysis/operating_states.py.

The four rules behind the arrows

- **Start-up only comes after a Shut-down.** It never comes from Hot standby or from power. The one exception is the very first file (2025-10-30), which begins in Idle.
- **Power only comes after Hot standby.** The last 15 minutes before load are always counted as Hot standby ("pre-load").
- **Hot standby means steady.** If the hot stack's temperature rises or falls by 5 °C per hour or more with no load, that is Thermal adjustment.
- **A paused heat-up is still Start-up.** On 17–18 March the heater was stopped overnight and restarted. That 11-hour pause stays inside the Start-up and is not counted as Idle or Shut-down.

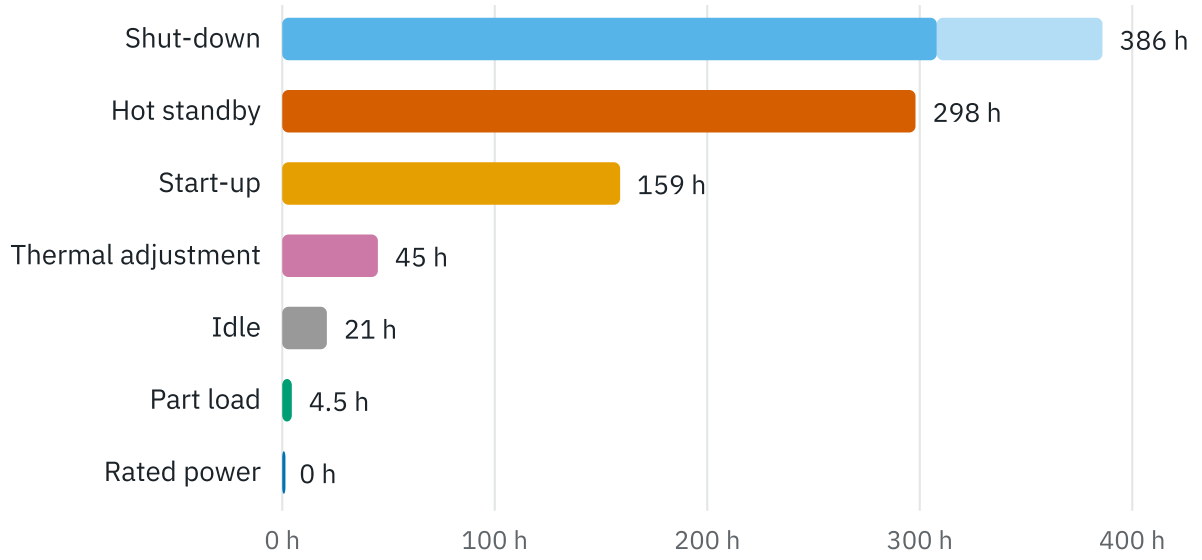
The next page describes each stage. The last page follows four real days through the map.

> **If you remember one thing:** heat up only after a stop, make power only after a steady hot hold. Every other move is a change of temperature or a stop.

PAGE 2 · WHAT EACH BOX MEANS

The seven stages

Each stage is recognised from just four signals: stack current, stack temperature and how fast it changes, the air heater, and the air flow.



Where the 914 hours went. The plant spent most of its time cooling down or held hot. It made power for only 4.5 hours, in 10 short runs. The Shut-down bar is split into Cooling (dark) and At ambient (light).

Shut-down

The plant takes the load off and cools the stack. It stays in Shut-down while waiting cold, until the next Start-up.

How we spot it	Load ramps to zero, then the temperature falls with air cooling or on its own. Once the stack is within 30 K of room temperature with everything off, the sub-stage is *At ambient*.
In the data	19 episodes · 386 h (Cooling 308 h · At ambient 78 h)

Start-up

The stack is heated from cold, or from warm, until it is hot and steady.

How we spot it	No load, temperature rising by more than 10 °C per hour, heater on. A heat-up that is paused and resumed counts as one Start-up.
In the data	15 episodes · 159 h

Hot standby

The stack is hot (about 510 °C or 575–610 °C) and held steady, ready to make power.

How we spot it	No load, temperature changing by less than 5 °C per hour. Also covers the 15 minutes just before load (pre-load).
In the data	26 episodes · 298 h

Thermal adjustment

The stack is already hot. With no load, the operators raise or lower its temperature.

How we spot it	No load, above 450 °C, temperature moving by 5 °C per hour or more: a re-heat or a cool-back.
In the data	18 episodes · 45 h

Part load

The stack makes power, but less than its rating.

How we spot it	Current between 1 A and 30 A per string. The highest seen was 16.2 A, 33.6 kW.
In the data	10 episodes · 4.5 h

Rated power

Full power: 50 kW AC, 30 A per string.

How we spot it	Never reached in the data.
In the data	0 episodes · 0 h

Idle

A data label, not a plant process: the plant is off and the stack is still warm, with no Shut-down before it.

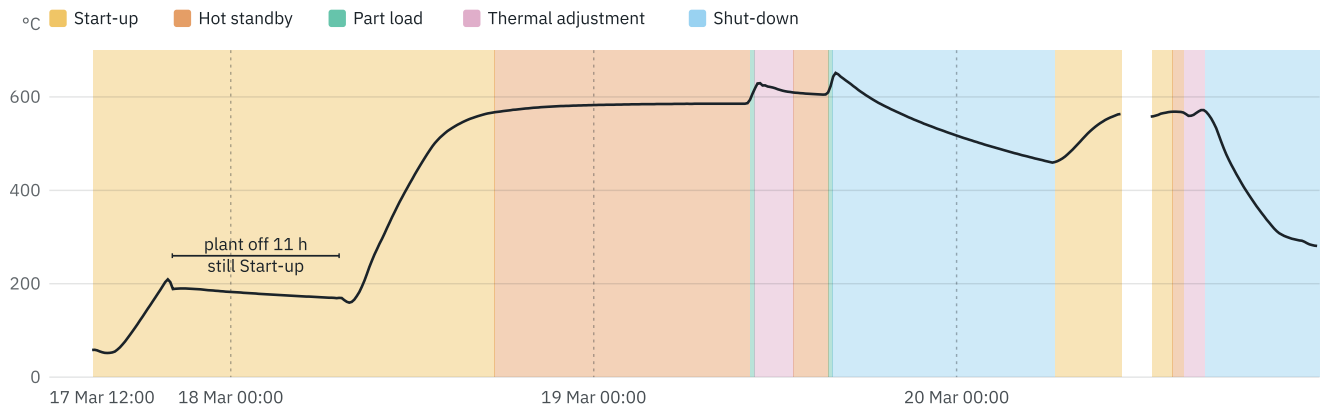
How we spot it	Only seen on 2025-10-30, the first file. Everything is off, and the stack is more than 30 K above room temperature.
In the data	2 episodes · 21 h

> **If you remember one thing:** power was made for only 4.5 of 914 hours. Almost all the data describes heating, holding and cooling.

PAGE 3 · THE MAP ON REAL DATA

A real week: 17–20 March 2026

Here are four days of stack temperature with each stage shaded. Press Play on the map page to follow the same days box by box.



Stack temperature (median of the stack thermocouples, 10-minute means). On the evening of 17 March the heater was stopped at about 190 °C. The plant stayed off for 11 hours, cooling by about 2 °C per hour, then the heat-up resumed. That pause stays part of the Start-up. The white gap on 20 March is missing logger data.

What happens, in order

- **Start-up**, 17 Mar 14:56 → 18 Mar 17:27: 58 → 567 °C, including the overnight pause.
- **Hot standby** overnight to 19 Mar 10:20, steady at 567–589 °C.
- **Part load** for 17 minutes, then **Thermal adjustment** (no load, re-heat and cool-back), then **Hot standby** again.
- **Part load** again, ended by a safety trip, then **Shut-down** with air cooling.
- On 20 March the stack is still at 459 °C, so a new **Start-up** heats it back. This is allowed because it follows a Shut-down.

Moves outside the design

Only three moves in 39 days break the map. All three are heat-ups that were stopped before the stack got hot.

Move	Times	When	Why
Start-up → Shut-down	2	2025-11-13, 2026-01-30	The heat-up was stopped before the stack was hot, and the plant cooled down.
Start-up → Idle	1	2025-10-30	The heat-up stopped at about 215 °C on the first day, and nothing came after it in the file.

Still open: should "heat-up stopped before Hot standby" become an allowed move, an *aborted start-up*?

> **If you remember one thing:** a real run zig-zags between Hot standby, Part load and Thermal adjustment, but it always enters through Start-up and leaves through Shut-down.